

CS 747? For undertones of intelligent ϵ learning Agents—

Post midsem topics.

Reinforcement Learning... Its a process!

what I already covered...

1) multi armed bandits.

regret, exploration vs exploitation.

UCB, KL-UCB, Thompson Sampling.

> How to act optimally when information is incomplete.

2) MDPs

• value functions

• Policy evaluation

• Policy iteration

• Value iteration } Bellman operator.

Q function:

$$Q(s, a)$$

> If model is known, planning is an optimisation
problem

From here-on - A) classic RL. lec 12-15

- Intro to RL
 - learning from interaction || unknown MDP.
 - Prediction vs control → meaning? •
- RL formalization -
 - Prediction problem || control problem... interesting •
 - Model based RL
 - optimal in the limit
- Monte Carlo methods... class of math?
 - MC predictions...
 - estimate values from complete episodes? ✓
 - sample for returns.

any other context? further down. ✓
- Temporal Difference learning...
 - maximum likelihood estimates } ??
 - least square estimates
 - MC vs TD comparison?

"important lecture in classical RL"

- Dynamic Programming — RL connection.
 - SARSA
 - Q-learning ! Hot topic
 - Function Approximation...
 - linear approximation
 - general approximation
 - Policy Gradient methods -
 - Direct policy optimisation.
 - REINFORCE } sounds exciting...
- slightly
nudge
breakdown.
- Visit Lec 16-20
once again.

- Batch RL ?
 - offline datasets.
 - experience replay
 - Fitted Q iteration.
 - Replay buffers in DQN.
 - Model based RL -
 - Dyna-Q } SOTA or legacy?
 - planning + Learning...
- "very relevant in 2026" ? really?

- Evolution & learning...
 - evolutionary algorithms. • sounds fancy...
 - Lamarckian vs Darwinian learning...
 - Baldwin effect. } whats dis!
 - Neuro evolution.
-

- Classical Search } whats the problem stmnt!
 - state-space search..
 - A^* search... was a big deal in RC.
 - Game tree and planning.
 - Monte Carlo tree search
 - UCT // like ucB. sorta...
 - Alpha Go
 - MCTS; Deep networks; Self-play
 - Integration of search + RL
-

ASK MORE, LEARN MORE.

Q) What's PPO?

≡ proximal policy optimisation...

- Policy gradient method...

learns policy directly.

$$\pi_{\theta}(a|s)$$

∴ optimise $J(\theta) = E[\text{return}]$

- classic algorithm REINFORCE.

- naturally handles continuous actions

means what?

action space is continuous?

or ~~are~~ follow up actions are?

- PPO constrains the gradient ascent...

clips the update ratio.

$$r_t(\theta) = \frac{\pi_{\theta}(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}$$

$r_t(\theta)$ is clipped into a trust region.

outcome:

stable

simple

"works surprisingly well"?

* Is the default RL algorithm for years.

Hm...

Q-learning

MC vs TD ✓

PPO ✓

DQN

Q) what's monte-carlo? context outside of RL?

"class of algorithms, that use random sampling
to estimate qty that are tough to compute".

... kinda lame if you ask me.

Monte-carlo $\rightarrow$ casino name; ref. to randomness

* General principle; need $E[f(x)]$

• tough to compute analytically.

• do sampling & approx as

$$\approx \frac{1}{N} \sum_i f(x_i)$$

* Used in ...

* Statistics, for expectation

* Bayesian Inference: MCMC

(markov chain monte carlo)

> when can't sample from complex distribution;
construct a markov chain whose stationary dist.
is the target distribution.

> Algorithms for this: metropolis-Hastings

Gibbs sampling

Hamiltonian Monte Carlo.

* in RL;

• we want $V(S)$;

• run episodes & get G_i sample return,

as one estimate...

$$V(S) \approx \frac{1}{N} \sum_i G_i$$

lol...

* why TD was invented?

temporal Difference...

• to avoid waiting till end of episode...

• use latest available val of next state.

BOOTSTRAPPING

Q) What's Q-Learning? And How done?

$Q(s, a) \rightarrow$ action value function.

$$Q(s, a) \rightarrow Q(s, a) + \alpha \left[\underbrace{r + \gamma \cdot \max_{a'} Q(s', a') - Q(s, a)}_{\text{TD error}} \right]$$

* maintain a Q-table; usually converges and approaches Bellman Optimality Equations.

* can choose action with ϵ -greedy.

* This is off-policy algorithm.

hence
off-policy.

Behaviour policy: policy that generates data.

Target policy: policy that you're trying to learn.

• if behaviour = target; then on-policy (PPO)

• if behaviour $\rightarrow$ noisy explorative,
 & we learn optimal policy; then off-policy

* Q-table approach; can't work with images
Robots

so use DQN to replace
table with NN. Continuous
state space;

